

A META-ANALYSIS OF THE RELATIONS AMONG TRAINING CRITERIA

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An augmented framework for training criteria based on Kirkpatrick's (1959a, 1959b, 1960a, 1960b) model divides training reactions into affective and utility reactions, and learning into post-training measures of learning, retention, and behavior/skill demonstration. A total of 34 studies yielding 115 correlations were analyzed meta-analytically. Results included substantial reliabilities across training criteria and reasonable convergence among subdivisions of criteria within a larger level. Utility-type reaction measures were more strongly related to learning or on-the-job performance (transfer) than affective-type reaction measures. Moreover, utility-type reaction measures were stronger correlates of transfer than were measures of immediate or retained learning. These latter findings support recent concurrent thinking regarding use of reactions in training (e.g., Warr & Bunce, 1995). Implications for choosing and developing training criteria are discussed.

Training researchers agree on the importance of evaluating training (e.g., Goldstein, 1993). There is equally strong agreement among training practitioners on the difficulty of doing so (Carnevale & Schulz, 1990). For any training evaluation to be valuable, however, training

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criteria must be psychometrically sound, meaningful to decision makers, and must be able to be collected within typical organizational constraints (Tannenbaum & Woods, 1992). Research has revealed that by far the most commonly collected training criteria are trainee reactions (Bassi, Benson, & Cheney, 1996; Saari, Johnson, McLaughlin, & Zimmerle, 1988) which, although easy to collect, may or may not be related to other, often more meaningful indicators for training evaluation.

Perhaps unsurprisingly, Kirkpatrick's four-level model (1959a, 1959b, 1960a, 1960b) continues to be the most prevalent framework for categorizing training criteria. This simple taxonomy of training criteria became very popular in business and academia because it addressed a need to understand training evaluation simply yet systematically (Shelton & Alliger, 1993). The model's simplicity is appealing but, as revealed in more recent work, this simplicity is also a liability. Alliger and Janak (1989) conducted a meta-analytic review of the literature based on Kirkpatrick's model. They concluded that:

[Kirkpatrick's model] provides a vocabulary and rough taxonomy for criteria. At the same time, Kirkpatrick's model, through its easily adopted vocabulary and a number of (often implicit) assumptions, can tend to misunderstandings and overgeneralizations (pp. 331-332).

Although there are problems with Kirkpatrick's model, just how best to think about training criteria is not clear. Perhaps Kirkpatrick's taxonomy requires revision. Some researchers have gone further and presented compelling arguments that entirely different and better models of training evaluation are needed (Holton, 1996; Kraiger, Ford, & Salas, 1993). New approaches are undoubtedly called for, and a thorough model of training *effectiveness* must include much more than is addressed by any taxonomy of training *criteria* (cf. Holton, 1996). Nonetheless, the Kirkpatrick typology remains by far the most influential and prevalent approach among practitioners (Kirkpatrick, 1996), and, to a certain extent, researchers. For this reason, it can still serve as a point of departure for communicating understandings about training criteria.

This article builds in part upon the meta-analytic work of Alliger and Janak (1989) in examining the relationship among training criteria. Since that article was published, training research has continued at an explosive pace (Ford, Kozlowski, Kraiger, Salas, & Teachout, 1997; Tannenbaum & Yukl, 1992). Training evaluation, as a topic, is constantly increasing in importance. Hence, understanding training evaluation criteria is also critical for researchers and practitioners alike. The current article furthers that understanding in several major ways. First, we augmented Kirkpatrick's typology to be less coarse, while maintaining a broad enough framework to facilitate a generalized understanding of

TABLE 1
Training Criteria Taxonomies

Kirkpatrick's taxonomy	Augmented framework
Reactions	Reactions Affective reactions Utility judgments
Learning	Learning Immediate knowledge Knowledge retention Behavior/skill demonstration
Behavior	Transfer
Results	Results

training evaluation results. Second, we conducted a meta-analysis of the relationships among training criteria based on this augmented model, using over twice as many studies and four times as many correlations as Alliger and Janak had available in 1989. Given the prevalence of trainee reaction measures in the workplace and in research, a central focus of interest was the relationship of trainee reactions with other criteria. For example, one question we address is, to what extent can trainee reactions be used as surrogate indicators of learning or behavior change? Third, because psychometric soundness is a requirement of any criterion, we examined the mean reliability of various training criteria. Fourth, we conducted a moderator analysis examining whether content similarity of criterion measures influences the relationship among them.

An Augmented Taxonomy

Below we briefly introduce a scheme for augmenting Kirkpatrick's taxonomy into one that is still simple but captures some additional important distinctions among criteria. Table 1 outlines the differences between the original and modified frameworks. The augmented framework guides the meta-analyses which follow. Please note that we do not propose the new framework as a comprehensive replacement for Kirkpatrick's original. Rather, the goal only was to provide an approach to coding for the current meta-analysis which was at least somewhat more sensitive to reasonable distinctions among training criteria.

Level 1. Reactions

Definition. Originally, to assess "reactions" was to ask trainees how they liked and felt about training. That is, reactions were emotionally

based opinions. Indeed, the term "reactions" seems to imply an immediate, more or less unthinking, response. However, the boundary between feeling and a more considered opinion is fuzzy at best, and trainers have asked a wide variety of "reaction" questions. Several researchers have suggested that reaction measures that directly ask trainees about the transferability or utility of the training should be more closely related to other criteria than would reactions measures that ask about "liking" (e.g., Alliger & Janak, 1989; Tannenbaum & Yukl, 1992). Attitude theorists acknowledge the difference between affective and more behaviorally evaluative responses (Eagly & Chaiken, 1992). Therefore, we have broken reactions into two basic components, affective and utility reactions. For the sake of the meta-analyses, we also needed a third sub-category which was a combination of the first two, because in many cases researchers combined both types of questions in one scale.

Interestingly, and independent of this research, Warr and Bunce (1995) suggested a tripartite division of reaction measures: enjoyment of training, usefulness of training, and difficulty of training. Warr and Bunce's (1995) first two concepts are our affect and utility reactions; the third concept, difficulty, is not captured in our scheme. In any case, training difficulty seems to be rarely asked of trainees.

Level 1a. Reactions as affect. Liking of training is easily the most common form of assessment of training. For example, "I found this training program to be enjoyable" is a typical reaction item. Usually such reactions are obtained via easy-to-administer post-training questionnaires. Reactions of trainees are extremely important for several reasons. First, trainees may be considered one of the "customers" of training. As such, assessment of their satisfaction with training seems entirely in keeping with most current models of provision of organizational services. Second, whether training is liked could have substantial influence on such distant variables as later training attendance, "word of mouth" advertising, subsequent training funding, and so forth.

Level 1b. Reactions as utility judgments. A second kind of reaction variable is made operational by asking such questions as "To what degree will this training influence your ability later to perform your job?" "Was this training job relevant?" and "Was the training of practical value?" These questions attempt to ascertain the perceived utility value, or usefulness, of training for subsequent job performance. An interesting question is whether answers to these utility questions correlate more or less strongly with later on-the-job application of trained skills or knowledge than do answers to affect-type reaction measures.

Level 1c. Combined reactions. In some cases correlations for reaction scales were available, but separate items for affective or utility reactions

were reported only as a combined score. In this case, a "combined reactions" category was coded for. It is interesting to note that when reaction scores are reported only in combined form, practioners and researchers may sometimes be acting on the assumption that all reactions tap a single underlying construct.

Level 2. Learning

Definition. Usually, learning as a training criterion is indexed by results of traditional tests of declarative knowledge. Many forms of knowledge assessment, however, could fit under this label. The assessment of mental models, for example, is a type of knowledge assessment, albeit relatively newer and much less common than typical measures of knowledge (Kraiger et al., 1993). Procedural knowledge, or performance of trained tasks immediately after training, also demonstrates learning. Hence, we include three subcategories of learning: knowledge that is assessed immediately after training (most common), knowledge that is assessed at a later time, and behavior/skill demonstration assessed immediately after training.

Level 2a. Immediate post-training knowledge. Immediate post-training assessment of learning is fairly common in the training literature. Usually knowledge is assessed by multiple choice test responses, answers to open-ended questions, listings of facts and so forth. That is, trainees are asked to indicate, in one of several ways, how much they know about the training topic(s). From our literature search we found overwhelming indication that traditional tests (e.g., multiple choice tests) are by far the most common, while newer methods of eliciting knowledge structure or associations are, as yet, virtually unused.

Level 2b. Knowledge retention. Sometimes training evaluators assess knowledge at a later time rather than (or in addition to) immediately after training. We coded such studies as "retention" measures of learning. That is, Level 2b measures are equivalent to Level 2a measures except that they are administered at some point later than just after training.

Level 2c. Behavior/skill demonstration. Kirkpatrick actually used the term "behavior" to refer to any behavioral changes that occur as a result of training. However, he did not make a clear distinction between behavior demonstrated in the training context and behavior demonstrated on the job. That is, his Level 3 may include both results of behavioral skill tests administered at the conclusion of training (i.e., indications of "can do") as well as on-the-job performance (i.e., indications of "does do"). It seems in keeping with Kirkpatrick's original intent, however, to retain Level 3 as representative of *transfer* of training to the job environment. Therefore, we include in Level 2c any indicators of behavioral

proficiency when these are measured within the training, rather than the work environment. Thus, in addition to simulations, such immediate post-training measures as behavioral role plays, behavioral reproduction, scores/grades in a performance-centered class, and ratings of training performance are typical of the kinds of measures that we classified here.

Level 3. Transfer

Definition. Although skilled performance that was assessed at the conclusion of training was coded as Level 2c, we term Level 3 "transfer," to emphasize the on-the-job nature of criteria in this category. A measure was classified as indicating on-the-job performance whenever it appeared that the measure was not only taken some time after training, but that it was in fact some measurable aspect of job performance. Sometimes ratings were used to indicate on-the-job performance; work samples, work outputs, and outcomes were also reported.

Note that although retained knowledge falls under Level 2, behavior that is retained and applied to the workplace is considered transfer, Level 3. This seems most in keeping with an important distinction: It is application to the job that, in most cases, defines training success (Alliger, Tannenbaum, & Bennett, 1995). Simple indication of retained knowledge may not.

Level 4. Results

Definition. "Results" criteria are meant to be those where organizational impact is indexed. Examples of results criteria include productivity gains, customer satisfaction, cost-savings, employee morale (for manager training), and profitability. In some ways, organizational results criteria represent, for training evaluation, the "ultimate" criteria (Brogden & Taylor, 1950). That is, they are at once the most distal from training, and often perceived as the most fundamental to judging training success. These qualities make results Level 4 training criteria seem highly desirable—and, indeed, when available, it may make good sense to collect them. However, organizational constraints greatly limit the opportunities for gathering Level 4 data (Shelton & Alliger, 1993; Tannenbaum & Woods, 1992), and most training efforts are incapable of directly affecting results level criteria. Unfortunately, the expectations of sponsors of training in regard to results-level outcomes may be unrealistic. Those charged with evaluation efforts may need to manage unrealistic sponsor expectations early in the evaluation process (Tannenbaum, 1996).

Another problem with Kirkpatrick's framework is that it is fairly vague about results level criteria. Some indicators such as employee attendance or scrappage rates could be categorized equally well as behavioral (Level 3) or results (Level 4) criteria. In any case, only three studies provided correlations that might be categorized as being based on Level 4 criteria, so we have not focused on this level in our current meta-analysis.

Content Overlap: Moderator Analysis

Fundamental differences between the different levels of criteria may explain the modest intercorrelations found in Alliger and Janak (1989) (e.g., learning tests are different from behavioral indicators). However, another explanation could be that different measures focused on different training content. For example, a learning measure might have focused on very specific aspects of handling hazardous materials as addressed in training, while a performance measure from the same study may have been a more global rating of safety behavior. In this case, a low correlation between the criteria may be attributable to low content overlap rather than a fundamental difference between learning and behavior. Therefore, in an attempt to better understand the relationship between criteria, we decided to conduct a moderator analysis based on the closeness of criteria to training content (that is, on the content validity of the measures). Each measure was rated as either being close to training content (i.e., highly content valid) or divergent from training content (less content valid). The prediction here is that if both measures are rated close to training content, then they would correlate more highly than if one or both measures were not constructed in such a way as to closely reflect training content.

Method

An updated manual search was conducted for each of the journals used in Alliger and Janak (1989; *Journal of Applied Psychology*, *Academy of Management Journal*, *Academy of Management Review*, *Journal of Applied Behavioral Science*, *Personnel Psychology*, *Personnel*, and *Training and Development Journal*). In addition, a manual search was conducted examining other journals reporting training research (*Group and Organizational Studies*, *Human Factors*, *Human Relations*, *International Journal of Man-Machine Studies*, *International Journal of Psychology*, *Journal of Management*, *Journal of Organizational Behavior*, *Public Personnel Management Journal*, and *Training and Education*).

A computer search was conducted using Psyclit and ERIC, databases that contain abstracts of psychological and educational research, as well as *Dissertation Abstracts International*. Direct communications were also made to researchers conducting training research in order to obtain unpublished studies or studies in press. Finally, references in review papers were also searched.

Alliger and Janak's (1989) meta-analysis produced 12 studies and 26 correlations. The current meta-analysis almost trippled the number of studies, providing a total of 34, and more than quadrupled the number of correlations, yielding 115.

The method of meta-analysis used was that developed by Hedges and Olkin (1985). Like other approaches, this method of research synthesis permits the generation of mean effect sizes and related confidence intervals, as well as tests of statistically significant differences between mean effect sizes. The effect size of interest, of course, for this article is the correlation coefficient. The Hedges/Olkin approach corrects effect size estimates for bias and uses inverse variance estimates, rather than sample size, for weighting purposes. Any bias is small, however, and resulting weighted mean correlations will be virtually identical to analogous estimates obtained via other meta-analytic procedures.

Results of Meta-Analysis

The results of the meta-analysis are presented in Table 2. An appendix including the citation for each study, the format of the measure, measure reliability, timing of measurement, class under each major level (e.g., "affective" under "reactions"), and coding of measure closeness to course content, including a description and/or example of each study measure is also available from the authors.

Reliabilities

Table 2 presents, on the diagonal, the results of a meta-analysis of criterion reliabilities. Average reliabilities for all categories of measures were above .58, with six out of seven above .75.

Correlations Involving Reactions

Affective and utility reactions tended to correlate positively, although only three studies provided such correlations. The overall average correlation between reactions of any type and immediate learning was only .07. This result corroborates Alliger and Janak's (1989) findings—

TABLE 2
Mean Sample-size Weighted Correlations Among Training Criteria

	Reactions				Learning									
	Affective <i>r</i>	<i>n</i>	Utility <i>r</i>	<i>n</i>	Combined <i>r</i>	<i>n</i>	Immediate <i>r</i>	<i>n</i>	Retained <i>r</i>	<i>n</i>	Behavior <i>r</i>	<i>n</i>	Transfer <i>r</i>	<i>n</i>
Reactions														
Affective	.82 (.81)	12	.34 (.28)	3			.02 (.01)	11			.03 (.01)	9	.07 (.03)	6
Utility			.86 (.85)	5			.26 (.20)	6			.03 (-.08)	3	.18 (.12)	3
Combined					.82 (.80)	5	.14 (.09)	6			.12 (.07)	8	.21 (.16)	9
Learning														
Immediate							.77 (.75)	14	.35 (.29)	2	.18 (.16)	13	.11 (.08)	16
Retained									.58 (.53)	2	.14 (.05)	2	.08 (.03)	4
Behavior											.85 (.84)	9	.18 (.11)	7
Transfer													.86 (.85)	13

Note: Values in parentheses show lower 95% confidence bound for mean correlation; *n* is number of studies combined in calculating each mean correlation. Empty cells indicate that one or fewer correlations were available. Reliabilities are on the diagonal.

training reactions should not be used blindly as a surrogate for the assessment of learning of training content. Affective reactions alone correlated on average just about zero with immediate learning. However, there were indications that utility reactions, considered alone, did correlate somewhat with immediate learning. As would also be predicted, those reactions that appeared to have both affective and utility characteristics correlated less than utility reactions alone, but more than affective reactions alone. The few reaction-retention correlations available provided an average index of relationship that again was very small. One advance of this study over Alliger and Janak (1989) is the separate examination of immediate post-training behavior/skill demonstration. On average, again, reactions did not correlate highly with this index of training effectiveness. An interesting result of this meta-analysis is that utility and combined reactions, as in the case of immediate and retained learning, again correlated moderately with transfer. Affective reactions, however, correlated less strongly with transfer.

Correlations Involving Learning (Other Than With Reactions)

Only two studies reported correlations between immediate and delayed learning measures. As reasonably expected, these correlations (which are a kind of index of reliability of learning over time) average moderately positive. Both immediate learning (2a) and retention (2b) correlated positively with immediate behavior/skill demonstration measures. The moderate positive linkage between knowledge and behavior/skill demonstration found by Alliger and Janak (1989) was confirmed. It is interesting to note that the size of relationship between post-training behavior/skill demonstration and on-the-job performance is similar to that for utility or combined reactions and on-the-job performance. Thus, given these results, behavioral learning predicts "transfer" no better than utility or combined reactions.

Correlations Involving Results

Immediate learning correlated substantially with results on average (.52). There were, however, only two studies reporting such correlations. Only one study reported a correlation between transfer and results.

Moderator Analysis: Closeness of Criteria to Training Content

A moderator analysis that examined closeness of the criteria to training content (content validity of measures) was carried out. Each mea-

sure was rated as either being close to training content (i.e., highly content valid) or divergent from training content (less content valid). The hypothesis was that if both measures are rated close to training content, then they would correlate more highly than if one or both measures were not constructed in such a way as to closely reflect training content. No correlations were coded divergent-divergent, because most measures reflected training content at some level. But, in accordance with prediction, the mean correlations between measures that were coded close-close was higher (mean weighted $r = .16$, based on 45 correlations) than those coded divergent-close (mean weighted $r = .04$, based on 42 correlations). The lower 95% confidence intervals for these two mean correlations do not include zero, and the difference between them is statistically significant ($p < .001$). However, it should be noted that *all* the measures coded divergent were reactions, so that level of criteria and content validity of criteria to training content were confounded. This means that the lower correlation for divergent-close could be due to the level of criteria rather than the moderator itself.

Discussion

One hundred and fifteen correlations were identified for inclusion in this meta-analysis. Although four times the amount uncovered in 1988, the number seems small given the expansiveness of the search. However, we have learned again what Alliger and Janak (1989) noted, namely, that even when studies clearly measured criteria on several levels, the inter-correlations among levels are often not noted in publications. Fortunately, this state of affairs seems to have improved somewhat, with more recent studies reporting such correlations. It may well be that in the past, the emphasis of training research was almost entirely on training effectiveness as indicated by mean change or mean group differences in criteria. Recently, however, there appears to be increasing awareness about understanding training effectiveness more broadly (e.g., Kraiger, et al., 1993), including the relationships among criteria.

The results reveal that at most, there are modest correlations between the various types of training criteria. Not surprisingly, the strongest correlations were exhibited between different criteria from within the same level, which may indicate convergent validity. Affective and utility reactions were correlated more strongly with each other (.34) than with other measures, and immediate and retained learning measures were correlated more strongly with each other (.35) than with other measures as well. This suggestion of convergent validity is tentative, however, because of the small number of correlations being combined.

Because reaction measures are the easiest to collect, it would be ideal

if they could be used as surrogate measures of learning and transfer. Interestingly, the magnitude of the relationship between training satisfaction and job performance is about the same as that typically exhibited between job satisfaction and job performance (Iaffaldano & Muchinsky, 1985).

As expected, utility and combined reactions correlated with on-the-job performance more highly than did affective reactions. This may be attributable to the more specific nature of the utility measures. The more behaviorally specific attitudes are, the more likely they are to predict behavior. Ajzen and Fishbein (1973) write, "While measures of attitude toward an object, such as obtained by the Thurstone, Likert, or semantic differential techniques, may perhaps be related to a person's general behavioral tendency with respect to the object, it appears that for the prediction of a given act, attitudinal variables more specific to the act in question will have to be considered" (p. 56). Weigel, Vernon, and Tognacci (1974) also found that the more specific the content of the attitude measure is to the behavioral criterion, the higher the relationship between the two. "Attitude measures should be expected to predict only behaviors that are appropriate to or specified by the attitude under consideration" (p. 728). Ajzen and Fishbein (1977) propose that low correlations may stem from the inconsistency between the levels of specificity of attitude and behavior variables. They reviewed a number of studies examining the attitude-behavior relationship in terms of the specificity or the congruence between target and action elements. The results tend to suggest "the relations between attitudes and behaviors tend to increase in magnitude as the attitudinal and behavioral entities come to correspond more closely in terms of their target and action elements" (p. 911). A recent meta-analysis by Kraus (1995) also found higher correlations between attitude and behavior when the two variables were measured at corresponding levels of specificity. Thus, utility reactions may in fact be more useful, for known theoretical reasons, for predicting on-the-job behavior than are typical affective measures. As one reviewer put it, perhaps "what we think is useful may correlate with what we use."

Although the utility reactions versus affective reactions results were anticipated, the utility versus learning results were not. Surprisingly, utility reactions exhibited higher correlations with on-the-job performance than did either immediate or retained measures of learning. One possible explanation assumes that trainees' utility reactions are influenced by their knowledge of the work environment to which they will return. Transfer (or "on-the-job performance") is a function of both knowledge acquisition and the favorability of the work environment for allowing new skills to be applied (Baldwin & Ford, 1988; Tracey, Tannenbaum,

& Kavanaugh, 1995). Objective measures of learning indicate whether trainees have acquired knowledge but do not capture whether they will be able to use it on the job. When asked to provide utility reactions, trainees may implicitly consider the situational constraints they will face when they return to the job (Peters, O'Connor, Eulberg, & Watson, 1988), as well as how much they believed they learned during training. A reviewer suggested that reactions may index "will," while learning indexes "skill." In any case, it needs to be stressed that many of the correlations in Table 2 are based on only a few studies, so that any firm conclusions about relative strength of transfer prediction by certain criteria (e.g., reactions vs. learning) are precluded.

Regardless of their relationship with other measures, from a pragmatic perspective, trainee reactions are important. Although positive reactions do not guarantee organizational support, negative reactions can often have an adverse effect on the training department. However, overall the results of this meta-analysis support Alliger and Janak's findings that reaction measures cannot be used as surrogates of other measures. In particular, affective reactions are unrelated to other indicators—liking does not equate to learning or to performing. If the purpose for collecting reaction measures is to predict or indicate transfer it would be best to ask utility-oriented questions. If both utility and affective questions are asked, it appears that these should be treated separately, with the utility questions being used to provide the better estimate of potential transfer.

With regard to reliability, three points can be made. First, training criteria, overall, seem to have reassuringly high reliabilities. A second point is that immediate measures have slightly higher reliability than more delayed measures of the same criterion. This makes sense if the training intervention is at its maximum general impact in the most immediate case (Bennett & Arthur, 1997). In other words, as time passes the intervention has differential impact on trainees, with some staying at the same level of proficiency or knowledge as immediately after training, while others gain or decrease in this regard. Third and finally, learning measure reliabilities are somewhat lower than either reaction or performance reliabilities. This may reflect the tendency for these measures to be more heterogeneous in content (i.e., they may cover a broader conceptual range) than the assessment of either reactions or performance. In any case, because one important criterion characteristic is reliability, it is reassuring to know that most training research criteria do possess that characteristic.

The results of the moderator analysis suggest that if training criteria do not overlap in content, convergence between or among them should not be expected. A lack of convergence, then, among criteria could

simply be a sign that the criteria were not, in fact, designed to cover the same aspects of training. Thus, depending on the goals of the evaluation, convergence might or might not be expected or even desirable. In some cases, it might be best to use multiple criteria with minimal overlap to get a more complete picture of the effect of training. The important point is for the researcher/evaluator to pursue a conscious criterion-development strategy vis-à-vis content validity; in this way, regardless of the relationships among criteria, the chance for misinterpreting those relationships is minimized.

Limitations and Future Research

One limitation of the current meta-analysis is the number of studies that fall within the domain. Although the total has grown significantly since 1988, there are still a number of cells for which there are few or no correlations. In particular, there are very few published studies based on Level 4 criteria. We believe this reflects the inherent difficulty of conducting Level 4 evaluations rather than a reporting bias. However, it does limit us from providing any useful information about the relationship of Level 4 criteria with other training criteria.

A second limitation is that the meta-analysis is based upon Kirkpatrick's model. As we noted earlier, this model has several shortcomings. Although the augmented framework may be an improvement, it cannot address all concerns. New taxonomic models capture recent developments from areas like cognitive psychology that are not addressed in Kirkpatrick's framework. Currently, Kirkpatrick's model remains the most prevalent, but as new taxonomic models, for example that of Kraiger et al. (1993), become more common, then the work from this study will need to be revisited.

We focused a great deal of attention on reactions because they are by far the most common criterion of training effectiveness. One area in which future research may prove fruitful is the examination of alternative methods of gathering reaction data. Reactions are typically gathered from training participants at the conclusion of training. However, trainees are not the only customers of training, nor is the conclusion of training necessarily the optimal time to collect reaction data. Trainees may not always be the most important or best judges of training effectiveness. Future research should examine the value of gathering utility reaction-type data from supervisors of training participants and other business leaders. Do their observations confirm the value or utility of the training? This is consistent with developments in the area of 360 degree feedback.

Affective reaction data is appropriately and easily gathered at the

conclusion of training. Trainees should be able to offer immediate judgments of whether they liked the training. However, utility reactions require trainees to speculate about the future usefulness of training. By gathering reaction data 1, 3, or 6 months after training, trainees will have experienced whether the training was in fact useful, and should be in a better position to judge the utility of the training. Future research should examine when best to collect reaction data.

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